Git Guide.

Every Git and GitHub answer in one place.

Ask in plain language, or paste the error Git printed.
Exact commands, a danger level for each, and its undo.

an undo for everything · works offline · no trackers · MIT

Amey Thakur

amey-thakur.github.io/GIT-GUIDE

A note from Amey

Git carries nearly every codebase on earth, and almost everyone who uses it learned it by accident: a command from a teammate, a midnight search, a ritual repeated without understanding.

Git Guide exists to end that way of learning. Ask in plain language, or paste the error Git printed, and get the exact commands with the two things most resources never give: how dangerous each one is, and how to take it back.

These pages hold the whole of it: the model that replaces memorizing, every command section, every error decoded, and the chapters on GitHub and on how teams work. The living version, searchable and always current, waits at **amey-thakur.github.io/GIT-GUIDE**.

Amey

Where your code lives

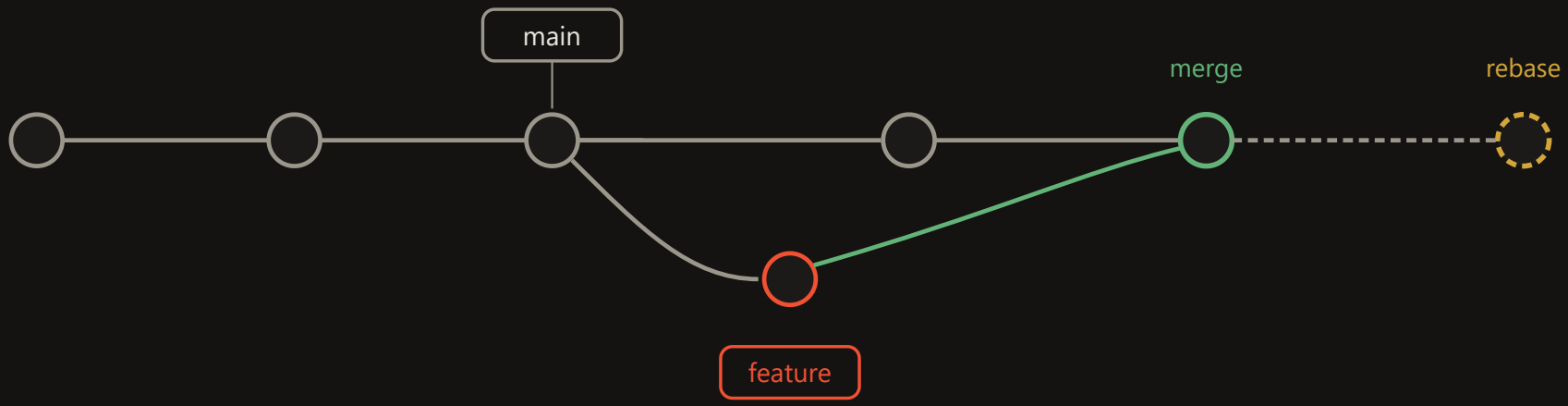
add copies a change into the staging area, the draft of your next commit. You choose what goes in.

commit seals the draft into a permanent snapshot in your local repository. Still only on your machine.

push uploads your commits; **fetch** downloads theirs and stops. **pull** is fetch plus merge.

restore copies recorded versions back over your files; **reset** moves the branch pointer itself. That is where history is un-happened.

Commits, branches, HEAD



A **branch** is a movable label on one commit. Creating one copies nothing; it is instant and free.

A **rebase** replays your commits as new ones for a straight line. It rewrites history: your own unshared work only.

A **merge** commit has two parents and joins the lines. History shows what truly happened; always safe on shared branches.

HEAD is where you stand, normally attached to a branch. Standing on a commit directly is detached HEAD: a state, not an error.

From zero to first push

1 Install

Windows: winget install Git.Git · macOS: brew install git · Debian: sudo apt install git

```
git --version
```

2 Identity

The same email as your GitHub account, or commits will not count on your profile

```
git config --global user.name "<name>" && git config --global user.email "<email>"
```

3 One key, every host

Add the public key to GitHub, GitLab, Bitbucket, Azure: the same key works on all

```
ssh-keygen -t ed25519 -C "<email>"
```

4 First repository

Joining: git clone <url> · Publishing a folder of yours:

```
gh repo create <name> --public --source . --push
```

5 The loop you will run forever

Edit, stage, commit, push. Everything else is a variation

```
git add <file> && git commit -m "<message>" && git push
```

6 When it goes wrong

It will, for everyone. Ask the site in plain language, or paste the exact error

```
amey-thakur.github.io/GIT-GUIDE
```

Setup and identity

`git config --global user.name "<name>"`
Name stamped into your commits

`git config --global user.email "<email>"`
Use your host account email so commits link to your profile

`git config --global init.defaultBranch main`
New repositories start on main

`git config --global core.editor "code --wait"`
Commit messages open in VS Code

`git config --global pull.ff only`
Pull never creates surprise merge commits

`git config --global fetch.prune true`
Deleted remote branches disappear locally too

`git config --global rebase.autoStash true`
Pull with rebase works even with edits in progress

`git config --global core.autocrlf true`
Windows line endings handled once; macOS and Linux use input

`git config --list --show-origin`
Every setting and the file it comes from

Start a project

`git init -b main`
Turn this folder into a repository

`git clone <url>`
Copy a repository, any host, full history

`git clone --depth 1 <url>`
Latest snapshot only, fastest

`git remote add origin <url>`
Connect a local repository to a host

`git push -u origin main`
First push; -u links the branches for good

The daily loop

`git status -sb`
What changed, one line per file

`git diff`
Unstaged edits, line by line

`git diff --staged`
What the next commit will contain

`git add <file>`
Stage a change; git add -p picks piece by piece

`git commit -m "<message>"`
Seal the staged snapshot

`git commit --amend --no-edit`
Add staged changes to the last commit

`git fetch`
See what the remote has before touching anything

`git pull --rebase`
Get remote commits, replay yours on top

`git push`
Publish your commits

Branches

`git switch -c <branch>`
Create and move in one step

`git switch <branch>`
Move to an existing branch

`git switch -`
Back to the branch you just left

`git branch --show-current`
Where am I

`git branch -vv`
Every local branch with its remote and ahead-behind state

`git branch -a`
Every branch, local and remote

`git push -u origin <branch>`
Publish a new branch

`git branch -d <branch>`
Delete a merged branch; -D forces

`git push origin --delete <branch>`
Delete it on the remote too

`git branch -m <old> <new>`
Rename

Merge and rebase

`git merge <branch>`
Bring a branch in; history stays true

`git merge --no-ff <branch>`
Always keep the feature visible as one bubble

`git rebase main`
Replay your branch on top of main

`git rebase -i HEAD~<n>`
Rewrite, reorder, squash your last n commits

`git cherry-pick <hash>`
Copy one commit onto this branch

`git merge --abort`
Back out of a conflicted merge

`git rebase --continue`
Resume after resolving a conflict

Undo and rescue

`git restore <file>`
Discard edits, back to last commit

`git restore --staged <file>`
Unstage, keep the edits

`git commit --amend -m "<new message>"`
Rewrite the last commit message

`git reset --soft HEAD~1`
Uncommit, keep everything staged

`git reset --hard HEAD~1`
Erase the last commit and its work

`git revert <hash>`
Undo by adding an opposite commit; safe when pushed

`git reflog`
Every state you have been in; the safety net

`git reset --hard ORIG_HEAD`
Undo the merge, rebase, or pull that just happened

`git clean -nd`
Preview deleting untracked files; -fd deletes

Stash

`git stash push -u -m "<label>"`
Set everything aside, untracked included

`git stash list`
What is shelved

`git stash pop`
Bring the newest back and drop it

`git stash apply stash@{<n>}`
Bring a specific one back, keep it shelved

`git stash show -p`
See inside before applying

Inspect and search

`git log --oneline --graph --all`
The whole history as a graph

`git log -p -- <file>`
Every change to one file; --follow crosses renames

`git log --grep="<text>"`
Find commits by message

`git log -S "<text>"`
Find commits that added or removed this text

`git diff main...<branch>`
What the branch changed since it split; the pull request view

`git shortlog -sn --all`
Commits per author, ranked

`git blame -w <file>`
Who last touched each line, ignoring whitespace

`git show <hash>`
One commit in full; --stat for files only

`git bisect start`
Binary-search history for the commit that broke it

`git branch -a --contains <hash>`
Which branches carry this commit

Tags and releases

`git tag -a v1.0.0 -m "<note>"`

Mark a release

`git push origin v1.0.0`

Tags do not push themselves

`git tag -d <tag>`

Delete locally; push origin --delete <tag> for the remote

`git describe --tags`

Nearest release to where you stand

`git switch --detach v1.0.0`

Visit a release; switch - to come back

Files and ignoring

`printf "node_modules/\n" >> .gitignore`

Ignore by pattern; affects untracked files only

`git rm --cached <file>`

Stop tracking, keep on disk

`git mv <old> <new>`

Rename and stage in one step

`git update-index --skip-worktree <file>`

Ignore your local edits to a tracked file

`printf "* text=auto\n" >> .gitattributes`

Line endings settled for the whole team, committed once

`touch <folder>/.gitkeep`

Keep an empty folder

Every platform, same Git

Git is identical everywhere; only the remote URL and sign-in differ. The same SSH public key works on every host.

`git clone git@github.com:<user>/<repo>.git`
GitHub over SSH

`git clone git@gitlab.com:<user>/<repo>.git`
GitLab

`git clone git@bitbucket.org:<workspace>/<repo>.git`
Bitbucket

`git clone https://dev.azure.com/<org>/<project>/_git/<repo>`
Azure DevOps (Azure Repos)

`git clone https://git-codecommit.<region>.amazonaws.com/v1/repos/<repo>`
AWS CodeCommit

`git config --global credential.helper manager`
Browser sign-in for HTTPS remotes, all hosts

`git remote set-url origin <url>`
Move between hosts or between HTTPS and SSH

`git clone --mirror <old> && git push --mirror <new>`
Migrate anywhere with full history

No host at all

Git predates the hosts and needs none of them.

`git init --bare ~/server/repo.git`
Your own remote on any machine or shared drive

`git bundle create repo.bundle --all`
The whole repository as one file: USB, email, air gap

`git clone repo.bundle -b main <folder>`
Clone from the file, no network

`git format-patch -3`
Last three commits as mailable patch files

`git am <file>.patch`
Apply mailed patches, authorship intact

`git archive --format=zip -o src.zip HEAD`
Clean source export, no .git

History surgery

Everything here rewrites history. Fresh clone first, backup kept, force-with-lease after, teammates re-clone.

`git commit --fixup <hash>`
Mark a fix as belonging to an earlier commit

`git rebase -i --autosquash <hash>^`
Melt fixups into their targets

`git filter-repo --invert-paths --path <file>`
Erase a file from all history

`git filter-repo --strip-blobs-bigger-than 10M`
Erase every giant blob

`git push --force-with-lease`
The only force push worth typing

Scale and speed

`git clone --filter=blob:none --sparse <url>`
Huge repository, download almost nothing

`git sparse-checkout set <folders>`
Materialize only what you work on

`git worktree add ../<repo>-review <branch>`
Second folder, second branch, zero stashing

`git lfs track "*.psd"`
Large binaries stored properly

`git count-objects -vH`
How heavy is this repository really

`git maintenance start`
Background upkeep keeps big repositories fast

Submodules

`git submodule add <url> <path>`
Pin another repository at an exact commit inside yours

`git clone --recurse-submodules <url>`
Clone a repo and its submodules in one step

`git submodule update --init --recursive`
Fill the empty folders after a plain clone

`git submodule update --remote`
Move pins to each submodule's latest commit

Quality of life

`git config --global help.autocorrect prompt`
git statsu asks: run git status instead?

`git config --global alias.st "status -sb"`
git st forever

`git config --global alias.lg "log --oneline --graph --all"`
git lg: the graph

`git config --global rerere.enabled true`
Never resolve the same conflict twice

`git config --global gpg.format ssh`
First of three lines to signed, Verified commits

`git commit --allow-empty -m "rerun"`
Retrigger CI without touching code

`git config core.hooksPath .githooks`
Versioned hooks the whole team shares

How teams run Git

Working alone

Commit at every working state; push is your backup; branch before experiments

```
git add -A && git commit -m "<message>" && git push
```

Feature branch

The team default: main stays releasable, every change arrives by pull request

```
git switch -c <change> && git push -u origin <change>
```

Fork

For repositories you cannot push to, which is all of open source

```
gh repo fork <owner>/<repo> --clone
```

Trunk-based

Branches live hours, not weeks; tiny changes merge daily behind flags

```
git config --global pull.rebase true
```

Releases and hotfixes

A release is a tag; a hotfix is a branch from that tag, merged back after shipping

```
git switch -c hotfix-2.1.1 v2.1.0
```

The rules that keep teams safe. Never rewrite a branch others build on; rewrite your own freely with force-with-lease. Pull with rebase. Protect main so nothing lands without review and green checks. When anything is unclear: git status names the state you are in, and usually the way out.

Your first repository

`Name`
Short, lowercase, hyphens: weather-app

`Add a README`
The front page, and your first commit

`Add .gitignore`
Pick your language; noise stays out from day one

`Choose a license`
MIT if unsure; none means all rights reserved

The pull request path

branch · commit · push · open the PR · checks run · review answers with commits · merge, and the branch retires.

`gh pr create --fill`
Open it from the terminal

`gh pr checkout <number>`
Review one locally

`gh pr merge --squash --delete-branch`
Land it clean

The words nobody explains

`repository` a project and its entire history; everyone says repo

`README.md` the front page, rendered automatically

`Markdown` plain text with light marks; GitHub renders it everywhere

`.gitignore` what Git must never track

`LICENSE` the legal terms of reuse

`public, private` everyone reads, only you write; or invitation only

`star` a bookmark and a thank-you; stars rank repositories

`watch` subscribe to issues, PRs, and releases

`clone vs fork` to your machine; to your account

`issue` a tracked conversation about one bug or request

`default branch` usually main: what visitors see, what PRs target

`gist` one shareable file with its own URL

`organization` a shared account for a team

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The errors you will meet

The message, then the cause. Every fix is one search away on the site.

`fatal: not a git repository (or any of the parent directories): .git`

This folder, and none above it, contains a .git directory. You are outside the project.

`remote: Repository not found.`

The URL has a typo, the repository is private and you lack access, or you are authenticated as the wrong account.

`git@github.com: Permission denied (publickey).`

The host never received a key it recognizes: no key loaded, key not added to your account, or an HTTPS problem disguised by an SSH URL.

`fatal: Authentication failed for 'https://github.com/...'`

The credential your machine sent is wrong, expired, or a password where a token is required.

`! [rejected] main -> main (non-fast-forward)`

The remote has commits you do not have. Git refuses to overwrite them.

`! [rejected] main -> main (fetch first)`

Same cause as non-fast-forward: someone pushed before you.

`error: src refspec main does not match any`

The branch you are pushing does not exist locally, almost always because there is no commit yet.

`fatal: refusing to merge unrelated histories`

The two sides share no common commit, typically a fresh git init pulled against a remote that already has commits.

`CONFLICT (content): Merge conflict in <file>`

Both sides changed the same lines; Git needs a human decision.

`error: Your local changes to the following files would be overwritten by merge`

The operation needs to rewrite files you have edited but not committed. Git refuses to destroy them silently.

`fatal: The current branch has no upstream branch`

This local branch has never been connected to a remote branch, so plain git push does not know where to go.

`You are in 'detached HEAD' state.`

Not an error. You checked out a commit or tag directly, so you stand on a snapshot instead of a branch.

The errors you will meet

The message, then the cause. Every fix is one search away on the site.

`fatal: unable to auto-detect email address`

Git has no identity to stamp into the commit.

`error: cannot pull with rebase: You have unstaged changes`

A rebase must start from a clean tree, and yours has edits.

`fatal: unable to access '...': Could not resolve host: github.com`

A network problem, not a Git problem: no internet, VPN or proxy in the way, or DNS down.

`fatal: unable to access '...': SSL certificate problem`

Something intercepts HTTPS, usually a corporate proxy with its own certificate that Git does not trust yet.

`error: RPC failed; HTTP 408 curl 22 (or 400, or curl 55)`

A large push choked, often against a proxy or server limit on request size.

`remote: error: GH001: Large files detected.`

GitHub refuses files over 100 MB in normal Git storage.

`fatal: Unable to create '.git/index.lock': File exists.`

Another Git process is running, or one crashed and left its lock behind.

`fatal: detected dubious ownership in repository`

The repository folder belongs to a different OS user than the one running Git, common in containers, WSL, and network drives.

`warning: LF will be replaced by CRLF the next time Git touches it`

Windows line endings meeting Unix line endings. Harmless, but configure it once and it goes quiet.

`error: pathspec '<name>' did not match any file(s) known to git`

The file or branch name does not exist as typed: a typo, wrong case, or a branch you have not fetched.

`fatal: a branch named '<name>' already exists`

You are creating a branch whose name is taken.

`error: the branch '<name>' is not fully merged`

git branch -d protects commits that exist nowhere else. This is the guardrail working.

The errors you will meet

The message, then the cause. Every fix is one search away on the site.

`fatal: destination path '<folder>' already exists and is not an empty directory`

git clone creates the folder, and a folder with that name is already there.

`error: remote origin already exists`

You are adding a remote named origin to a repository that already has one.

`fatal: 'origin' does not appear to be a git repository`

You pushed or pulled against a remote name that is not configured here.

`Your branch and 'origin/main' have diverged`

Both you and the remote gained different commits since you last synced.

`Your branch is ahead of 'origin/main' by N commits`

Not an error. You have local commits the remote lacks; push when ready.

`fatal: bad object HEAD (or: loose object is corrupt)`

The object database is damaged, usually a crash, a full disk, or a synced-folder service fighting over .git.

`error: unable to create file <path>: Filename too long`

Windows' 260-character path limit, hit by deep folder structures.

`error: invalid path '<name>'`

The repository contains a file name Windows forbids: reserved names like aux or con, or characters like colon and question mark, committed from Linux or macOS.

`warning: You appear to have cloned an empty repository.`

Not an error. The remote exists but has no commits yet.

`fatal: this operation must be run in a work tree`

You are inside the .git folder itself, or in a bare repository that has no working files by design.

`fatal: tag '<name>' already exists`

Tags are unique names; this one is taken.

`error: cannot spawn .git/hooks/pre-commit: No such file or directory`

A hook script is missing, not executable, or has Windows line endings breaking its shebang line.

The errors you will meet

The message, then the cause. Every fix is one search away on the site.

`error: gpg failed to sign the data`

Signing is switched on but the signing setup is incomplete or the key is unavailable.

`error: Pulling is not possible because you have unmerged files.`

An earlier merge stopped at conflicts and is still waiting for you to finish it.

`hint: You have divergent branches and need to specify how to reconcile them.`

Newer Git asks you to choose a pull strategy once instead of silently merging.

`fatal: could not read Username for 'https://github.com': No such device or address`

Git wanted to ask for credentials but no terminal was attached, the classic CI and script failure.

`fatal: early EOF (the remote end hung up unexpectedly)`

A large clone or fetch died mid-transfer: unstable connection, proxy timeout, or a server cutting long downloads.

`xcrun: error: invalid active developer path (/Library/Developer/CommandLineTools)`

A macOS update removed the command line tools, and git on a Mac lives inside them.

`Could not open a connection to your authentication agent.`

ssh-add needs the ssh-agent process, and none is running in this shell.

`fatal: Not possible to fast-forward, aborting.`

Your pull is set to fast-forward only, and the branches have diverged, so a decision is needed.

`You are not currently on a branch.`

Pull needs a branch to update, and you stand on a detached commit.

`remote: Permission to <repo> denied to github-actions[bot].`

A workflow tried to push, but Actions runs with read-only permissions unless the workflow asks for more.

The living version

Eight doors, one address: **amey-thakur.github.io/GIT-GUIDE**

Finder
ask anything, or paste the error

Start
zero to first push

Learn
the model, step by step

Fix
guided rescue

Errors
Git's messages, decoded

GitHub
first repo to branch protection

Workflows
how teams run Git

Cheatsheet
everything, one line each

Every answer also lives in the repository's Discussions; a question the guide cannot answer becomes its next addition. MIT licensed, no trackers, works offline.